Capstone Manuscript Content and Outline Evaluation

Web Based Service Management System with Data Analytics Using Machine Learning Algorithms for Shoelotskey Villamor Pasay

Angeles Gravino Sedanto

Researchers	Angeles, Charmaine Rose G. Gravino, Kylane C. Sedanto, Melody D.
Evaluation scope	Content accuracy, completeness, research-method alignment, evidence, and compliance with the prescribed NAAP BSIT/BSIS Capstone Layout. Formatting is excluded.
Overall content rating	84.0% Major Revision Required
Outline compliance	97% Substantially compliant, with ordering and placement corrections

Overall finding: The manuscript is unusually detailed and provides substantially better technical evidence than a perception-only capstone evaluation. It includes historical-data auditing, baseline comparisons, a train-validation-test design, manual-versus-digital measurements, objective technical tests, ISO/IEC 25010 expert assessment, TAM, and a complete Chapter V. Nevertheless, the machine-learning claim still requires major revision because the operational model was trained and tested on simulated records generated from verified business rules. Its near-perfect performance demonstrates that the model learned the simulated rule environment; it does not yet demonstrate reliable prediction of actual Shoelotskey completion times.

Rating Summary

Chapter	Rating	Assessment
Chapter I Introduction	82%	Complete and well aligned, but the prescribed order is not followed and the ML claim needs clearer qualification.
Chapter II Review of Related Literature and Studies	80%	Relevant thematic coverage and synthesis are present; source verification, critical comparison, and real-world ML validation literature need strengthening.
Chapter III System Analysis and Design	91%	All major prescribed topics are present and technically detailed. Minor internal inconsistencies and stronger traceability are still required.
Chapter IV Research Methodology Results and Discussion	79%	Strong objective testing and transparent limitations, but the simulated-data model cannot establish real operational predictive validity. Reliability and comparison issues remain.
Chapter V Summary Conclusions and Recommendations	88%	Complete, cautious, and generally evidence-aligned, but continued operational use should not yet be recommended as if the predictive component were validated on real completion data.
OVERALL	84.0%	Major Revision Required before final acceptance of the predictive claim.

Chapter I Introduction

1 Project Context

STATUS: PRESENT BUT MISPLACED AND PARTLY DUPLICATED

My evaluation/comment: The manuscript begins with several pages of introductory discussion, a Problem Statement, a pre-assessment summary, and Research Questions before the formal Project Context heading. These opening paragraphs already perform the function of the Project Context. Under the prescribed outline, Project Context should be the first identified topic. The current arrangement creates duplication and makes the official outline appear out of sequence.

My suggested correction: Move and consolidate the opening discussion, the strongest baseline findings, and the institutional/business context under Project Context. Place the Problem Statement and Research Questions after Scope and Limitations only if the adviser requires them as additional sections; otherwise integrate the problems into Project Context and align the research questions with the objectives.

> **Sample revision:** *Shoelotskey Villamor-Pasay relies on paper records and separate files for job orders, service-status updates, inventory, and report preparation. The pre-assessment conducted from March 13 to 15, 2026 found an estimated job-order retrieval time of 3 to 10 minutes, a 2- to 3-day delay in reflecting service-status updates, and 1 to 2 hours for manual report preparation. These findings establish the operational basis for developing a centralized service-management and analytics system.*

2 Problem Statement and Pre Assessment

STATUS: PRESENT AND EVIDENCE BASED BUT REQUIRES SOURCE CLARIFICATION

My evaluation/comment: The section is stronger than a purely narrative problem statement because it provides dates, respondents, methods, estimated times, transaction volume, and evidence sources. However, several values are described as estimates from the owner or staff while other values are presented as observations or available records. The evidence type must be separated so that recalled estimates are not treated as timed measurements.

My suggested correction: Label each baseline as owner estimate, staff estimate, direct timed observation, or documentary count. State the number of observations and records examined. If the values will be used in the system comparison, repeat the measurements under a controlled protocol.

> **Sample revision:** *The 3- to 10-minute retrieval range was reported by the owner and staff during the pre-assessment and is treated as a perceived baseline. A separate controlled retrieval test involving 10 matched tasks was conducted for the formal manual-versus-digital comparison reported in Chapter IV.*

3 Project Purpose and Description

STATUS: PRESENT AND GENERALLY CORRECT

My evaluation/comment: The section clearly distinguishes descriptive analytics, Random Forest prediction, and rule-based computation. This distinction is methodologically important. However, the phrase machine learning prediction may still lead readers to assume real-world validity even though the deployed model was developed from simulated operational data.

My suggested correction: State prominently that the ML Predicted Release Date is a prototype advisory output trained on simulated operational data and that the Official Expected Release Date remains rule-based until sufficient real Ready-for-Release records become available.

> **Sample revision:** *The predictive component is a prototype advisory model developed from a simulated operational dataset based on verified service configurations. It is not a substitute for the Official Expected Release Date and will require external validation using actual Ready-for-Release records before operational reliance is recommended.*

4 Contextual Framework

STATUS: PRESENT BUT MISNAMED

My evaluation/comment: The manuscript uses the heading Conceptual Framework, while the prescribed NAAP outline requires Contextual Framework. The IPO framework with Evaluation and Feedback is clearly explained and appropriately separates operational processes from evaluation frameworks, but the required heading must still be followed for outline compliance.

My suggested correction: Rename the subsection Contextual Framework. Retain the IPO content, but change any output phrased as improved performance into a measured outcome unless Chapter IV provides the corresponding evidence. Clearly mark simulated-data ML evaluation as prototype validation.

> **Sample revision:** *The contextual framework uses an Input-Process-Output model with Evaluation and Feedback to show how business records and operational requirements are transformed into system outputs and how the developed system is assessed using technical tests, ISO/IEC 25010, TAM, and machine-learning performance measures.*

5 General Objective

STATUS: PRESENT BUT SHOULD QUALIFY THE MACHINE LEARNING COMPONENT

My evaluation/comment: The general objective is aligned with system development and operational improvement. The objective should not imply that the current ML model has already improved completion-date prediction in actual operations.

My suggested correction: Use design, development, implementation, and evaluation language, and distinguish operational system outcomes from prototype predictive-model outcomes.

6 Specific Objectives

STATUS: PRESENT MEASURABLE AND WELL ALIGNED

My evaluation/comment: The objectives are among the strongest portions of Chapter I because they specify comparison indicators, regression metrics, ISO/IEC 25010:2023 characteristics, TAM constructs, and recommendations. Objective 2, however, should explicitly say that the finalized model is evaluated on simulated operational data, while real-world external validation remains future work.

My suggested correction: Revise Objective 2 to identify Dataset A and Dataset B and prevent the simulated test result from being read as real operational accuracy.

> **Sample revision:** *Evaluate the feasibility and limitations of the historical Order-to-Claim target using actual records, and evaluate the prototype Random Forest Regressor for Received-to-Ready-for-Release duration using a separately partitioned simulated operational dataset, while clearly limiting conclusions to simulated conditions.*

7 Hypothesis

STATUS: NOT APPLICABLE BUT SHOULD BE ACKNOWLEDGED

My evaluation/comment: The study uses descriptive comparisons and predictive performance measures without inferential hypothesis testing. The prescribed outline requires a hypothesis only when applicable.

My suggested correction: Add a short statement that no hypothesis was formulated because the study is developmental and descriptive. If inferential comparisons are later added, state the corresponding null and alternative hypotheses.

8 Scope and Limitations

STATUS: PRESENT DETAILED AND APPROPRIATELY CAUTIOUS

My evaluation/comment: The section correctly identifies the system boundary, excluded integrations, privacy limitations, historical-target problem, simulated dataset, small evaluator groups, and short pilot. This is a major strength. The most important limitation should be moved earlier and stated more directly: simulated performance cannot establish actual release-date prediction.

My suggested correction: Add the simulation-generation procedure, assumptions, service-duration distributions, noise process, and rules used to generate Dataset B. Explain that training and testing on partitions of the same synthetic generator assess internal consistency, not external validity.

Chapter I Rating: 82%

Chapter II Review of Related Literature and Studies

1 Thematic Review

STATUS: PRESENT AND RELEVANT

My evaluation/comment: The review covers service management systems, analytics, machine learning, Random Forest regression, MSMEs, software quality, and user acceptance. The sources are generally connected to the proposed system. The discussion should more consistently compare datasets, target variables, sample sizes, evaluation methods, and limitations rather than cite high performance values as general support for Random Forest.

My suggested correction: For every predictive study, identify the target, dataset source, number of records, split method, baseline, and external-validity limitation. Verify every reference against its original source and remove any source that cannot be traced or does not support the attached claim.

2 Machine Learning and Synthetic Data Literature

STATUS: PRESENT BUT INSUFFICIENT FOR THE FINAL MODEL DESIGN

My evaluation/comment: The literature supports regression and model comparison, but it needs a stronger discussion of synthetic-data validation. A model evaluated on data generated from the same operational rules can obtain very high performance because the data-generating relationship is stable and known. This is different from performance on actual operations, where unrecorded workload, technician availability, rework, material shortages, and unexpected delays affect completion time.

My suggested correction: Add literature on synthetic-data utility, simulation-to-real validity, data leakage, concept drift, external validation, temporal validation, and prediction intervals. Explain why actual Ready-for-Release records remain necessary.

3 ISO IEC 25010 and TAM Literature

STATUS: PRESENT AND ALIGNED

My evaluation/comment: The manuscript uses the nine quality characteristics associated with ISO/IEC 25010:2023 and treats expert perceptions separately from objective technical tests. TAM is also limited to user acceptance. This separation is correct.

My suggested correction: Cite the exact ISO edition consistently and avoid calling questionnaire ratings proof of conformity. Clarify that Actual System Use measured after a two-week mixed actual/simulated pilot is short-term self-reported use rather than sustained adoption.

4 Synthesis

STATUS: PRESENT AND GENERALLY STRONG

My evaluation/comment: The synthesis connects the reviewed technologies to the research design and distinguishes historical Dataset A from simulated Dataset B. It should more explicitly conclude that the literature supports feasibility but does not validate the local model.

My suggested correction: End the synthesis with a concise evidence gap describing the absence of a locally validated prediction model based on actual Ready-for-Release timestamps.

5 Research Gap

STATUS: EMBEDDED BUT NOT SEPARATELY IDENTIFIED

My evaluation/comment: A separate Research Gap heading is not required by the prescribed outline, but the gap is distributed across the review and synthesis. A concise, separately identified research gap would improve traceability.

My suggested correction: State that the gap concerns an integrated small-business service-management platform and the absence of validated real operational completion-time data, not merely the absence of a Random Forest model.

Chapter II Rating: 80%

Chapter III System Analysis and Design

1 Technical Background

STATUS: PRESENT AND TECHNICALLY DETAILED

My evaluation/comment: The stack, architecture, analytics, security, and machine-learning design are described. Maintain one authoritative description of the active model and dataset.

2 Research and Development Methodology

STATUS: PRESENT

My evaluation/comment: Waterfall is identified and the phases are documented. Ensure that each phase has a dated deliverable and approval evidence.

3 Software Development Life Cycle Model

STATUS: PRESENT

My evaluation/comment: The selected Waterfall model is appropriate for a defined single-client workflow, although model iteration should be documented when the prediction target changed.

4 Development Phases

STATUS: PRESENT

My evaluation/comment: The phases are detailed. Add a traceability matrix linking objectives, requirements, modules, test cases, and results.

5 Methodology Justification

STATUS: PRESENT

My evaluation/comment: The justification is acceptable. Explain how the target-variable revision was governed within a Waterfall process.

6 System Development Framework

STATUS: PRESENT

My evaluation/comment: The framework is present. Verify consistency with the Chapter I conceptual framework and avoid duplicating the same diagram under different labels.

7 Analysis of the Existing System

STATUS: PRESENT

My evaluation/comment: The existing workflow is described and supported by pre-assessment evidence.

8 Current Evaluation Procedures

STATUS: PRESENT

My evaluation/comment: The current procedures are discussed. Keep estimates separate from timed observations.

9 Limitations and Problem Analysis

STATUS: PRESENT

My evaluation/comment: The problem analysis is specific and operationally relevant.

10 Requirements Gap Analysis

STATUS: PRESENT

My evaluation/comment: The gap analysis is present. Add acceptance criteria and source evidence for every requirement.

11 Alternative Solutions and Feasibility Analysis

STATUS: PRESENT

My evaluation/comment: Alternatives are compared. Selection criteria should include privacy, maintainability, cost, offline failure, and predictive-data availability.

12 Proposed Technology Alternatives

STATUS: PRESENT

My evaluation/comment: Technology alternatives are discussed. Ensure the final selection exactly matches the implemented stack.

13 Feasibility Study

STATUS: PRESENT WITH ALL REQUIRED SUBTOPICS

My evaluation/comment: Technical, operational, economic, and schedule feasibility are all present.

14 Cost Benefit Analysis

STATUS: PRESENT AND COMPREHENSIVE

My evaluation/comment: The required cost and benefit components are included. Verify labor-rate, hosting, maintenance, useful-life, and benefit assumptions using documented bases. Avoid monetizing benefits that were not measured.

15 System Requirements Specification

STATUS: PRESENT

My evaluation/comment: Stakeholders, functional requirements, and non-functional requirements are present. Convert qualitative terms into measurable acceptance criteria.

16 Proposed System Architecture and Design

STATUS: PRESENT

My evaluation/comment: Architecture and modules are identified. Mark the rule-based Official Expected Release Date and prototype ML Predicted Release Date as separate decision paths.

17 Process Modeling and System Workflow

STATUS: PRESENT WITH ALL REQUIRED MODELS

My evaluation/comment: Context Diagram, DFD Level 0, DFD Level 1, and procedural workflow are present. Verify balancing, names, and exception flows.

18 Data Modeling and Database Design

STATUS: PRESENT WITH ALL REQUIRED MODELS

My evaluation/comment: Conceptual Data Model, ERD, and Logical Database Design are present. This fully satisfies the required outline.

19 System Security and Data Privacy Design

STATUS: PRESENT AND SUPPORTED BY TESTS

My evaluation/comment: Access control, encryption, privacy, and audit logging are discussed. Functional security tests are useful but should not be described as a penetration test or full security certification.

20 Implementation Environment and Technology Stack

STATUS: PRESENT WITH ALL REQUIRED SUBTOPICS

My evaluation/comment: Software, hardware, development tools, and deployment environment are present. Record exact versions and the tested production configuration.

Major Chapter III Technical Correction Synthetic Dataset Validity

STATUS: REQUIRES MAJOR REVISION

My evaluation/comment: Dataset B contains 10,000 simulated orders generated from verified service configurations and operational rules. The model uses service quantities, rush status, and calendar variables to predict a target produced within that simulated environment. The resulting R^2 of 0.9976 and 97.20% within ±1 day are therefore measures of performance against the synthetic generator. They do not demonstrate that the model can predict actual Ready-for-Release duration. The exceptional performance also warrants a generator audit for indirect target leakage or a nearly deterministic rule relationship.

My suggested correction: Document the complete synthetic-data generator, all distributions, duration rules, noise terms, workload assumptions, and random seeds. Have an independent reviewer inspect whether any feature or derived feature reveals the target-generating rule. Most importantly, collect actual Ready-for-Release timestamps and perform a temporal external validation. Until then, label the integrated output as a simulated-data prototype prediction.

> **Sample revision:** *The prototype model achieved high internal performance on a held-out partition produced by the same synthetic data-generating process. This result establishes consistency within the simulation only and must not be interpreted as evidence of real-world predictive accuracy. External temporal validation will be conducted after sufficient actual Ready-for-Release observations have been accumulated.*

Internal Text and Cross Reference Errors

Observed issue	Required correction
The sentence "Your manuscript already establishes this target" appears in the Chapter IV discussion.	Delete it. This is editorial or AI-assistance residue and is not manuscript prose.
The sentence "Your existing methodology documents..." appears in the Random Forest discussion.	Rewrite in third-person academic form and remove direct address.
The fragment "analytics_dashboard_reproducibi..." appears in the dashboard methodology.	Remove the filename or incomplete artifact reference and cite the appendix/evidence file formally if retained.
The enhancement discussion refers to Table 24 although the suggestions are in Table 42.	Correct the cross-reference to Table 42.

Observed issue	Required correction
The pre-assessment heading spells the business name as "SHOLOTSKEY."	Correct it to "SHOELOTSKEY."
"As shown in the Table 15, The..." contains an article and capitalization error.	Revise to "As shown in Table 15, the..."
The heading "Results and Discussions" is plural.	Use "Results and Discussion" to match the prescribed outline.
A subsection appears as ".4.3 Machine Learning Model Evaluation."	Correct the numbering and verify all heading numbers automatically.

Chapter III Rating: 91%

Chapter IV Research Methodology Results and Discussion

1 Research Design

STATUS: PRESENT AND APPROPRIATE

My evaluation/comment: The developmental-descriptive design is appropriate. The study correctly avoids inferential claims and uses separate technical, perception-based, and predictive evidence. The design should explicitly classify the ML evaluation as simulation-based prototype validation.

My suggested correction: Add one sentence distinguishing actual operational evaluation, historical target assessment, and synthetic model development.

2 Data Gathering Technique

STATUS: PRESENT DETAILED AND REPRODUCIBLE

My evaluation/comment: Dates, participants, interview duration, focus group, observation, document review, testing, pilot period, and evaluator groups are identified. This is strong. The manual retrieval and report tests need exact task lists, who performed each task, device conditions, and raw timing records in an appendix.

My suggested correction: Attach the protocols, anonymized raw timing logs, test scripts, and evidence references. Distinguish actual pilot activities from simulated pilot activities in the results.

3 Research Instruments and Validation

STATUS: PRESENT BUT RELIABILITY IS NOT ESTABLISHED

My evaluation/comment: The instruments underwent expert content validation, and the CVI method is described. However, a perfect I-CVI of 1.00 for 14 broad validation criteria demonstrates agreement about the instrument as a whole only if those criteria were actually applied item by item. It does not replace construct reliability. The manuscript explicitly states that Cronbach's alpha was not conducted.

My suggested correction: Report the item-level validation matrix for each questionnaire item, not only 14 general validation criteria. Conduct pilot reliability testing with respondents similar to the target users when feasible and report Cronbach's alpha or another appropriate reliability estimate per construct. If the sample remains too small, retain the limitation and avoid claiming established reliability.

> **Sample revision:** *The instruments demonstrated expert-assessed content validity. Internal consistency reliability was not established in the present study; therefore, the multi-item construct results are interpreted cautiously and are not presented as psychometrically validated scales.*

4 Statistical Treatment of Data

STATUS: PRESENT BUT THE LIKERT INTERVALS REQUIRE CORRECTION OR JUSTIFICATION

My evaluation/comment: The manuscript reports means, standard deviations, time reduction, MAE, RMSE, R^2, and tolerance accuracy. The four-point interpretation ranges are unequal: 1.00–1.50, 1.51–2.50, 2.51–3.50, and 3.51–4.00. This makes the extreme categories narrower than the middle categories and classifies 3.50 as Acceptable.

My suggested correction: Use equal-width intervals, normally 1.00–1.75, 1.76–2.50, 2.51–3.25, and 3.26–4.00, unless NAAP has an approved institutional interpretation scheme. Recalculate all verbal interpretations after selecting the final legend. Also clarify whether reported SD is the average item SD or the SD of respondent-level construct scores.

5 Population and Sampling Technique

STATUS: PRESENT AND INTERNALLY CONSISTENT

My evaluation/comment: Total population sampling for six intended users and purposive sampling for five IT experts are properly distinguished. The small groups are appropriate for client-specific formative evaluation but limit generalizability and psychometric analysis.

My suggested correction: List the qualification criteria of the IT experts and confirm that maintainability reviewers inspected technical artifacts. State that the six users are the complete intended-user population, not a statistically representative sample of MSMEs.

6 Manual Versus Digital Comparison

STATUS: PRESENT BUT NOT FULLY MATCHED

My evaluation/comment: The retrieval comparison uses measured tasks and reports mean and median times. The manuscript appropriately admits that manual and digital accuracy evidence came from different periods and denominators. Therefore, the result supports a descriptive operational comparison but not a controlled causal claim about accuracy.

My suggested correction: Use the same set of job orders and the same task definitions for both methods. Report the number of trials, operators, raw times, errors, missing records, incomplete records, and pass/fail criteria. If historical paper and new digital records remain unmatched, describe the accuracy finding as a record audit rather than a system comparison.

7 Dashboard Reporting Efficiency

STATUS: PRESENT BUT THE 99.9968 PERCENT CLAIM IS NOT AN END TO END COMPARISON

My evaluation/comment: The manuscript transparently states that manual time measured human preparation while digital time measured API response after the data had already been encoded. The caveat is correct, but the comparison still combines different task boundaries. A 0.0838-second API response cannot be directly compared with a complete 43.71-minute manual reporting workflow as if both represent the same process.

My suggested correction: Report two separate measures: application response time and end-to-end user task time. For an end-to-end comparison, begin from the same request and dataset and end when a verified report is available to the user. Retain the API benchmark as a technical performance measure, not the primary operational improvement percentage.

> **Sample revision:** *The dashboard endpoint generated the tested report data in a mean of 0.0838 second under the specified local test conditions. This is an application-response benchmark and is not directly equivalent to the complete manual report-preparation workflow. A separate end-to-end task comparison is required to quantify operational time savings.*

8 Historical Machine Learning Evaluation Dataset A

STATUS: PRESENT TRANSPARENT AND METHODOLOGICALLY SOUND FOR DIAGNOSING THE TARGET PROBLEM

My evaluation/comment: The manuscript correctly shows that the historical Random Forest did not outperform the existing business rules and that a negative R^2 means the model did not improve upon the reference variance explanation. Retaining the poor result is academically responsible. The 119 records and 26-record temporal holdout are too small for strong generalization, but they validly demonstrate that Claimed Date is a poor proxy for actual service completion.

My suggested correction: Retain this analysis. Emphasize that it is a target-validity diagnosis, not evidence supporting the deployed predictive model.

9 Operational Machine Learning Evaluation Dataset B

STATUS: PRESENT BUT NOT REAL WORLD VALIDATION

My evaluation/comment: The 70/15/15 split, model comparison, hyperparameter search, frozen final test set, baselines, residual analysis, and tolerance results are well documented. However, all partitions originate from the same synthetic generator. The test set is independent from model fitting but not independent from the underlying data-generating assumptions. The near-perfect scores may reflect rule reconstruction rather than prediction of real operational uncertainty.

My suggested correction: Do not use "independent test" without qualifying it as an independent synthetic test partition. Add a simple rule-based baseline derived from the generator itself. Compare the ML model against that baseline. The definitive validation must use later, untouched, actual Ready-for-Release records from a different time period.

> **Sample revision:** *On the held-out synthetic test partition, the prototype Random Forest reproduced the simulated operational patterns with MAE = 0.0815 day and R^2 = 0.9976. These values are internal simulation results and do not establish equivalent accuracy on actual service orders.*

10 ISO IEC 25010 Evaluation

STATUS: PRESENT AND CORRECTLY SEPARATES PERCEPTION FROM TECHNICAL EVIDENCE

My evaluation/comment: Five IT experts evaluated the nine ISO/IEC 25010:2023 characteristics, and objective tests were reported separately. The manuscript correctly states that the findings are not certification. This is a strong approach. Some test results may still be developer-executed and require independent verification.

My suggested correction: Identify the tester for every technical test, preserve screenshots/logs, add predetermined acceptance criteria, and have at least one qualified independent IT expert reproduce critical security, backup, recovery, concurrency, and compatibility tests.

11 Technology Acceptance Model

STATUS: PRESENT BUT LIMITED TO SHORT TERM SELF REPORTED USE

My evaluation/comment: TAM is administered to the full six-person intended-user population after a two-week pilot. The manuscript appropriately warns that Actual System Use does not prove long-term adoption. Scores of 4.00 with SD 0.00 should be interpreted cautiously because of the very small, client-dependent group and possible social desirability.

My suggested correction: Supplement self-reported Actual System Use with system logs showing login frequency, completed tasks, active days, errors, and continued use. Extend observation beyond two weeks if long-term adoption is claimed.

12 Results and Discussion

STATUS: PRESENT COMPREHENSIVE BUT CONTAINS EDITORIAL RESIDUE

My evaluation/comment: The chapter connects most results to objectives and includes limitations. Remove direct-address editorial phrases, incomplete filenames, incorrect table references, and numbering errors identified in this evaluation. These residues undermine confidence in an otherwise careful manuscript.

My suggested correction: Conduct a complete line-by-line cleanup and cross-reference audit. Verify that every table number cited in the discussion matches the actual table and that no drafting instructions remain.

Chapter IV Rating: 79%

Chapter V Summary Conclusions and Recommendations

1 Summary

STATUS: PRESENT COMPLETE AND OBJECTIVE ALIGNED

My evaluation/comment: The summary is organized according to the objectives and reports the central quantitative results. It appropriately distinguishes Dataset A from Dataset B and acknowledges the simulated nature of the final model. The phrase "achieved the machine learning objective" should be qualified as an internal simulation objective, not real operational validation.

My suggested correction: State separately what was demonstrated with actual data, what was demonstrated with simulated data, and what remains unvalidated.

2 Conclusions

STATUS: PRESENT GENERALLY CAUTIOUS AND SUPPORTED

My evaluation/comment: The conclusions correctly acknowledge unmatched observation periods, simulated data, limited respondents, and a short pilot. The conclusion that the system supports its intended operational context is reasonable for the non-predictive modules. The model conclusion must remain limited to simulated conditions.

My suggested correction: Avoid describing the integrated ML component as operationally effective until it passes external temporal validation on actual Ready-for-Release records. The rule-based Official Expected Release Date should remain the operational output.

3 Recommendations

STATUS: PRESENT RELEVANT BUT CONTINUED USE MUST BE CONDITIONED

My evaluation/comment: The recommendations appropriately call for actual Ready-for-Release data, external validation, longer use, security maintenance, automated testing, and future algorithm comparison. Recommending continued system use is acceptable for centralized records and reporting, but the predictive module must remain advisory and should be monitored for error.

My suggested correction: Add a controlled implementation condition: specify who reviews model errors, when retraining is allowed, the minimum real dataset, acceptance thresholds, rollback to the rule-based estimate, and protection against automatic deployment of an unvalidated model.

> **Sample revision:** *Shoelotskey may continue using the validated record-management, inventory, service-status, and reporting functions during a monitored pilot. The ML Predicted Release Date shall remain advisory. The Official Expected Release Date shall continue to follow approved business rules until the prototype model is extemally validated using an adequate temporal dataset of actual Ready-for-Release records and meets predetermined error thresholds.*

Chapter V Rating: 88%

Complete Outline Compliance Check

Prescribed topic	Status	Required action
Chapter I Project Context	Present but misplaced	Move to the beginning and consolidate duplicated opening discussion.
Project Purpose and Description	Present	Qualify the prototype ML output.
Contextual Framework	Present but mislabeled	Rename Conceptual Framework to Contextual Framework to follow the prescribed outline.
Objectives General and Specific	Present	Qualify simulated-data objective.
Hypothesis if applicable	Not applicable	Add a non-applicability statement.
Scope and Limitations	Present	Add full synthetic-generator limitations.

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Prescribed topic	Status	Required action
Chapter II Thematic RRL and Studies	Present	Strengthen critical comparison and source verification.
Synthesis	Present	End with an explicit local evidence gap.
Chapter III Technical Background	Present	Maintain one authoritative technical specification.
R and D Methodology SDLC Phases Justification	Present	Add requirement and test traceability.
System Development Framework	Present	Verify alignment and avoid duplicate frameworks.
Existing System Analysis and all five subtopics	Present	Keep estimated and measured baselines separate.
Feasibility Study all four areas	Present	Verify evidence and assumptions.
Cost Benefit Analysis all required components	Present	Verify rates, sources, and benefits.
SRS Stakeholders Functional Non Functional	Present	Make every NFR measurable.
Architecture Overview and Modules	Present	Separate rule based and ML outputs.
Context Diagram DFD 0 DFD 1 Workflow	Present	Audit balancing and exceptions.
Conceptual Data Model ERD Logical Design	Present	Fully compliant.
Security and Privacy all subtopics	Present	Require independent reproduction of critical tests.
Implementation Environment all subtopics	Present	Add exact versions and tested configuration.
Chapter IV Research Design	Present	Classify the ML design as simulation based.
Data Gathering Technique	Present	Attach raw protocols and logs.
Statistical Treatment	Present	Correct or justify unequal Likert intervals.
Population and Sampling	Present	Appropriate but limited generalizability.
Results and Discussion	Present	Remove drafting residue and qualify simulation results.
Chapter V Summary	Present	Separate actual and simulated evidence.
Conclusions	Present	Limit model conclusion to simulation.
Recommendations	Present	Condition continued use and deployment.

Priority Revisions Before Final Defense

1. Reposition Project Context as the first prescribed topic and remove duplication with the opening pages.
2. Revise the title, objective, or claim so the predictive component is clearly identified as a prototype trained and tested on simulated operational data.
3. Document the complete Dataset B generation process and audit it for direct or indirect target leakage.
4. Collect actual Ready-for-Release timestamps and conduct temporal external validation before claiming real-world predictive accuracy.
5. Compare the ML model with the exact rule-based generator or strongest operational business-rule baseline.
6. Report end-to-end report-generation task time separately from API response time; do not treat 99.9968% as a full workflow improvement.

7. Correct or justify the unequal four-point Likert interpretation ranges and recalculate interpretations if changed.
8. Provide item-level instrument-validation evidence and establish reliability when feasible; otherwise retain a clear limitation.
9. Supplement self-reported Actual System Use with objective system-usage logs and a longer observation period.
10. Remove editorial or AI-assistance residue, incomplete filenames, wrong table references, and numbering errors.
11. Have independent qualified testers reproduce critical security, backup, recovery, compatibility, concurrency, and performance tests.
12. Revise Chapter V after all corrected Chapter IV values and claims are finalized.

Final Rating and Panel Comment

Overall Content Rating: 84.0%
Outline Compliance Rating: 97%
Decision: Major Revision Required

My overall adviser comment: I find the manuscript substantially complete, technically detailed, and highly compliant with the prescribed capstone outline. The researchers have made a commendable effort to separate user perception, software-quality assessment, technical testing, historical-data analysis, and machine-learning evaluation. The transparent reporting of the poor historical Random Forest result and the limitation of Claimed Date as a target is a particular methodological strength. However, the active predictive model remains the central issue. Its very high performance was obtained from training, validation, and test partitions generated by the same simulated operational process. This shows performance within the simulation but does not yet establish reliable prediction of actual Shoelotskey completion times. I therefore require the researchers to qualify the title and conclusions, fully document and audit the synthetic-data generator, and complete external temporal validation using actual Ready-for-Release records before presenting the model as operationally accurate. The manuscript must also correct the Chapter I sequence, the Likert interpretation ranges, the unmatched workflow comparison, the instrument-reliability limitation, and all editorial residue and incorrect cross-references. Subject to these major revisions, the work has strong potential and is substantially more evidence-based than a perception-only system evaluation.

Reviewed and evaluated by:

ASSOC. PROF. MARY ANN F. ABALLIAR-VISTA, MEAM

Review Panel